

Cosmos SDK *Story*

HALBORN

Cosmos SDK - Story

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Date of Engagement by: October 29th, 2024 - November 12th, 2024

Summary

100% ⓘ OF ALL REPORTED FINDINGS HAVE BEEN ADDRESSED

ALL FINDINGS	CRITICAL	HIGH	MEDIUM	LOW	INFORMATIONAL
2	0	1	0	1	0

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1. Introduction

Story engaged Halborn to conduct a security assessment on their Cosmos SDK, beginning on October 29, 2024, and ending on October 15, 2024. The security assessment was scoped to cover their `cosmos-sdk` GitHub repository, located at <https://github.com/piplabs/cosmos-sdk/tree/> with commit ID `3d7ade1`.

2. Assessment Summary

The team at Halborn was provided two weeks for the engagement and assigned one full-time security engineer to assess the security of the Cosmos project. The security engineer is a blockchain and smart-contract security expert with advanced penetration testing, smart-contract hacking, and deep knowledge of multiple blockchain protocols.

The purpose of this assessment is to:

- Ensure that the **Golang components** operate as intended.
- Identify potential security issues.
- Identify lack of best practices within the codebase.
- Identify systematic risks that may pose a threat in future releases.

In summary, Halborn identified some improvements to reduce the likelihood and impact of risks, which should be addressed by the **Story team**. The main ones were the following:

- Changed `UnbondingIDKey` to a different key to not overlap with other keys, which would allow storage corruption in certain situations.
- Changed the duration types to the correct ones, as they did not matched the specs.

3. Test Approach And Methodology

Halborn performed a combination of manual and automated security testing to balance efficiency, timeliness, practicality, and accuracy in regard to the scope of the custom modules. While manual testing is recommended to uncover flaws in logic, process, and implementation; automated testing techniques help enhance coverage of structures and can quickly identify items that do not follow security best practices. The following phases and associated tools were used throughout the term of the assessment :

- Research into architecture and purpose.
- Static Analysis of security for scoped repository, and imported functions. (e.g., `staticcheck`, `gosec`...)
- Manual Assessment for discovering security vulnerabilities on the codebase.
- Ensuring the correctness of the codebase.
- Dynamic Analysis of files and modules in scope.

4. RISK METHODOLOGY

Every vulnerability and issue observed by Halborn is ranked based on **two sets of Metrics** and a **Severity Coefficient**. This system is inspired by the industry standard Common Vulnerability Scoring System.

The two **Metric sets** are: **Exploitability** and **Impact**. **Exploitability** captures the ease and technical means by which vulnerabilities can be exploited and **Impact** describes the consequences of a successful exploit.

The **Severity Coefficients** is designed to further refine the accuracy of the ranking with two factors: **Reversibility** and **Scope**. These capture the impact of the vulnerability on the environment as well as the number of users and smart contracts affected.

The final score is a value between 0-10 rounded up to 1 decimal place and 10 corresponding to the highest security risk. This provides an objective and accurate rating of the severity of security vulnerabilities in smart contracts.

The system is designed to assist in identifying and prioritizing vulnerabilities based on their level of risk to address the most critical issues in a timely manner.

4.1 EXPLOITABILITY

ATTACK ORIGIN (AO):

Captures whether the attack requires compromising a specific account.

ATTACK COST (AC):

Captures the cost of exploiting the vulnerability incurred by the attacker relative to sending a single transaction on the relevant blockchain. Includes but is not limited to financial and computational cost.

ATTACK COMPLEXITY (AX):

Describes the conditions beyond the attacker’s control that must exist in order to exploit the vulnerability. Includes but is not limited to macro situation, available third-party liquidity and regulatory challenges.

METRICS:

EXPLOITABILITY METRIC (M_E)	METRIC VALUE	NUMERICAL VALUE
Attack Origin (AO)	Arbitrary (AO:A) Specific (AO:S)	1 0.2

EXPLOITABILITY METRIC (M_E)	METRIC VALUE	NUMERICAL VALUE
Attack Cost (AC)	Low (AC:L)	1
	Medium (AC:M)	0.67
	High (AC:H)	0.33
Attack Complexity (AX)	Low (AX:L)	1
	Medium (AX:M)	0.67
	High (AX:H)	0.33

Exploitability E is calculated using the following formula:

$$E = \prod m_e$$

4.2 IMPACT

CONFIDENTIALITY (C):

Measures the impact to the confidentiality of the information resources managed by the contract due to a successfully exploited vulnerability. Confidentiality refers to limiting access to authorized users only.

INTEGRITY (I):

Measures the impact to integrity of a successfully exploited vulnerability. Integrity refers to the trustworthiness and veracity of data stored and/or processed on-chain. Integrity impact directly affecting Deposit or Yield records is excluded.

AVAILABILITY (A):

Measures the impact to the availability of the impacted component resulting from a successfully exploited vulnerability. This metric refers to smart contract features and functionality, not state. Availability impact directly affecting Deposit or Yield is excluded.

DEPOSIT (D):

Measures the impact to the deposits made to the contract by either users or owners.

YIELD (Y):

Measures the impact to the yield generated by the contract for either users or owners.

METRICS:

Impact Metric (M_I)	Metric Value	Numerical Value
Confidentiality (C)	None (I:N)	0
	Low (I:L)	0.25
	Medium (I:M)	0.5
	High (I:H)	0.75
	Critical (I:C)	1
Integrity (I)	None (I:N)	0
	Low (I:L)	0.25
	Medium (I:M)	0.5
	High (I:H)	0.75
	Critical (I:C)	1
Availability (A)	None (A:N)	0
	Low (A:L)	0.25
	Medium (A:M)	0.5
	High (A:H)	0.75
	Critical (A:C)	1
Deposit (D)	None (D:N)	0
	Low (D:L)	0.25
	Medium (D:M)	0.5
	High (D:H)	0.75
	Critical (D:C)	1
Yield (Y)	None (Y:N)	0
	Low (Y:L)	0.25
	Medium (Y:M)	0.5
	High (Y:H)	0.75
	Critical (Y:C)	1

Impact I is calculated using the following formula:

$$I = max(m_I) + \frac{\sum m_I - max(m_I)}{4}$$

4.3 SEVERITY COEFFICIENT

REVERSIBILITY (R):

Describes the share of the exploited vulnerability effects that can be reversed. For upgradeable contracts, assume the contract private key is available.

SCOPE (S):

Captures whether a vulnerability in one vulnerable contract impacts resources in other contracts.

METRICS:

SEVERITY COEFFICIENT (<i>C</i>)	COEFFICIENT VALUE	NUMERICAL VALUE
Reversibility (<i>r</i>)	None (R:N)	1
	Partial (R:P)	0.5
	Full (R:F)	0.25
Scope (<i>s</i>)	Changed (S:C)	1.25
	Unchanged (S:U)	1

Severity Coefficient *C* is obtained by the following product:

$$C = rs$$

The Vulnerability Severity Score *S* is obtained by:

$$S = min(10, EIC * 10)$$

The score is rounded up to 1 decimal places.

SEVERITY	SCORE VALUE RANGE
Critical	9 - 10
High	7 - 8.9
Medium	4.5 - 6.9
Low	2 - 4.4

SEVERITY	SCORE VALUE RANGE
Informational	0 - 1.9

5. SCOPE

FILES AND REPOSITORY

(a) Repository: [cosmos-sdk](#)

(b) Assessed Commit ID: [c553dc6](#)

(c) Items in scope:

- [x/auth/ante/ante.go](#)
- [x/auth/ante/ante_test.go](#)
- [x/auth/ante/basic.go](#)
- [x/auth/ante/basic_test.go](#)
- [x/auth/ante/expected_keepers.go](#)
- [x/auth/ante/ext.go](#)
- [x/auth/ante/ext_test.go](#)
- [x/auth/ante/fee.go](#)
- [x/auth/ante/feegrant_test.go](#)
- [x/auth/ante/fee_test.go](#)
- [x/auth/ante/setup.go](#)
- [x/auth/ante/setup_test.go](#)
- [x/auth/ante/sigverify_benchmark_test.go](#)
- [x/auth/ante/sigverify.go](#)
- [x/auth/ante/sigverify_test.go](#)
- [x/auth/ante/testutil/expected_keepers_mock.go](#)
- [x/auth/ante/testutil_test.go](#)
- [x/auth/ante/validator_tx_fee.go](#)
- [x/auth/autocli.go](#)
- [x/auth/client/cli/broadcast.go](#)
- [x/auth/client/cli/decode.go](#)
- [x/auth/client/cli/encode.go](#)
- [x/auth/client/cli/encode_test.go](#)
- [x/auth/client/cli/query.go](#)
- [x/auth/client/cli/query_test.go](#)
- [x/auth/client/cli/tx_multisign.go](#)
- [x/auth/client/cli/tx_sign.go](#)
- [x/auth/client/cli/tx_simulate.go](#)
- [x/auth/client/cli/validate_sigs.go](#)
- [x/auth/client/testutil/helpers.go](#)
- [x/auth/client/tx.go](#)
- [x/auth/client/tx_test.go](#)
- [x/auth/codec/bech32_codec.go](#)
- [x/auth/exported/exported.go](#)

- [x/auth/keeper/account.go](#)
- [x/auth/keeper/deterministic_test.go](#)
- [x/auth/keeper/genesis.go](#)
- [x/auth/keeper/grpc_query.go](#)
- [x/auth/keeper/grpc_query_test.go](#)
- [x/auth/keeper/keeper_bench_test.go](#)
- [x/auth/keeper/keeper.go](#)
- [x/auth/keeper/keeper_test.go](#)
- [x/auth/keeper/migrations.go](#)
- [x/auth/keeper/msg_server.go](#)
- [x/auth/keeper/msg_server_test.go](#)
- [x/auth/migrations/legacytx/amino_signing.go](#)
- [x/auth/migrations/legacytx/codec.go](#)
- [x/auth/migrations/legacytx/stdsign.go](#)
- [x/auth/migrations/legacytx/stdsignmsg.go](#)
- [x/auth/migrations/legacytx/stdsig_test.go](#)
- [x/auth/migrations/legacytx/stdtx_builder.go](#)
- [x/auth/migrations/legacytx/stdtx.go](#)
- [x/auth/migrations/legacytx/stdtx_test.go](#)
- [x/auth/migrations/v1/types.go](#)
- [x/auth/migrations/v2/store.go](#)
- [x/auth/migrations/v2/store_test.go](#)
- [x/auth/migrations/v3/store.go](#)
- [x/auth/migrations/v3/store_test.go](#)
- [x/auth/migrations/v4/migrate.go](#)
- [x/auth/migrations/v4/migrate_test.go](#)
- [x/auth/migrations/v5/migrate.go](#)
- [x/auth/migrations/v5/migrate_test.go](#)
- [x/auth/module.go](#)
- [x/auth/module_test.go](#)
- [x/auth/posthandler/post.go](#)
- [x/auth/signing/adaptor.go](#)
- [x/auth/signing/adaptor_test.go](#)
- [x/auth/signing/handler_map.go](#)
- [x/auth/signing/sign_mode_handler.go](#)
- [x/auth/signing/sig_verifiable_tx.go](#)
- [x/auth/signing/verify.go](#)
- [x/auth/simulation/genesis.go](#)
- [x/auth/simulation/genesis_test.go](#)
- [x/auth/simulation/proposals.go](#)
- [x/auth/simulation/proposals_test.go](#)
- [x/auth/testutil/app_config.go](#)
- [x/auth/testutil/expected_keepers_mock.go](#)
- [x/auth/testutil/util.go](#)

- [x/auth/tx/adapter.go](#)
- [x/auth/tx/aux_test.go](#)
- [x/auth/tx/builder.go](#)
- [x/auth/tx/builder_test.go](#)
- [x/auth/tx/config/config.go](#)
- [x/auth/tx/config/expected_keepers.go](#)
- [x/auth/tx/config.go](#)
- [x/auth/tx/config_test.go](#)
- [x/auth/tx/decoder.go](#)
- [x/auth/tx/direct_aux.go](#)
- [x/auth/tx/direct_aux_test.go](#)
- [x/auth/tx/direct.go](#)
- [x/auth/tx/direct_test.go](#)
- [x/auth/tx/encode_decode_test.go](#)
- [x/auth/tx/encoder.go](#)
- [x/auth/tx/legacy_amino_json.go](#)
- [x/auth/tx/legacy_amino_json_test.go](#)
- [x/auth/tx/query.go](#)
- [x/auth/tx/service.go](#)
- [x/auth/tx/sigs.go](#)
- [x/auth/tx/sigs_test.go](#)
- [x/auth/tx/testutil/expected_keepers_mock.go](#)
- [x/auth/tx/testutil/suite.go](#)
- [x/auth/tx/textual.go](#)
- [x/auth/types/account.go](#)
- [x/auth/types/account_retriever.go](#)
- [x/auth/types/account_retriever_test.go](#)
- [x/auth/types/account_test.go](#)
- [x/auth/types/auth.pb.go](#)
- [x/auth/types/codec.go](#)
- [x/auth/types/credentials.go](#)
- [x/auth/types/credentials_test.go](#)
- [x/auth/types/expected_keepers.go](#)
- [x/auth/types/genesis.go](#)
- [x/auth/types/genesis.pb.go](#)
- [x/auth/types/genesis_test.go](#)
- [x/auth/types/keys.go](#)
- [x/auth/types/params.go](#)
- [x/auth/types/params_legacy.go](#)
- [x/auth/types/params_test.go](#)
- [x/auth/types/permissions.go](#)
- [x/auth/types/permissions_test.go](#)
- [x/auth/types/querier.go](#)
- [x/auth/types/query.go](#)

- [x/auth/types/query.pb.go](#)
- [x/auth/types/query.pb.gw.go](#)
- [x/auth/types/tx.pb.go](#)
- [x/auth/vesting/autocli.go](#)
- [x/auth/vesting/client/cli/tx.go](#)
- [x/auth/vesting/client/cli/tx_test.go](#)
- [x/auth/vesting/exported/exported.go](#)
- [x/auth/vesting/module.go](#)
- [x/auth/vesting/msg_server.go](#)
- [x/auth/vesting/msg_server_test.go](#)
- [x/auth/vesting/testutil/expected_keepers_mock.go](#)
- [x/auth/vesting/types/codec.go](#)
- [x/auth/vesting/types/constants.go](#)
- [x/auth/vesting/types/expected_keepers.go](#)
- [x/auth/vesting/types/genesis_test.go](#)
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- [x/auth/vesting/types/period.go](#)
- [x/auth/vesting/types/test_common.go](#)
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- [x/auth/vesting/types/vesting_account.go](#)
- [x/auth/vesting/types/vesting_account_test.go](#)
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- [x/authz/authorization_grant_test.go](#)
- [x/authz/authorizations.go](#)
- [x/authz/authz.pb.go](#)
- [x/authz/client/cli/tx.go](#)
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- [x/authz/codec.go](#)
- [x/authz/errors.go](#)
- [x/authz/event.pb.go](#)
- [x/authz/expected_keepers.go](#)
- [x/authz/generic_authorization.go](#)
- [x/authz/generic_authorization_test.go](#)
- [x/authz/genesis.go](#)
- [x/authz/genesis.pb.go](#)
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- [x/authz/keeper/genesis_test.go](#)
- [x/authz/keeper/grpc_query.go](#)
- [x/authz/keeper/grpc_query_test.go](#)
- [x/authz/keeper/keeper.go](#)
- [x/authz/keeper/keeper_test.go](#)

- [x/authz/keeper/keys.go](#)
- [x/authz/keeper/keys_test.go](#)
- [x/authz/keeper/migrations.go](#)
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- [x/authz/keeper/msg_server_test.go](#)
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- [x/authz/migrations/v2/keys.go](#)
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- [x/authz/module/abci_test.go](#)
- [x/authz/module/autocli.go](#)
- [x/authz/module/module.go](#)
- [x/authz/messages.go](#)
- [x/authz/messages_test.go](#)
- [x/authz/proto_desc.go](#)
- [x/authz/query.pb.go](#)
- [x/authz/query.pb.gw.go](#)
- [x/authz/simulation/decoder.go](#)
- [x/authz/simulation/decoder_test.go](#)
- [x/authz/simulation/genesis.go](#)
- [x/authz/simulation/genesis_test.go](#)
- [x/authz/simulation/operations.go](#)
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- [x/authz/testutil/expected_keepers_mock.go](#)
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- [x/circuit/keeper/genesis_test.go](#)
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- [x/circuit/keeper/msg_server.go](#)
- [x/circuit/keeper/msg_server_test.go](#)
- [x/circuit/keeper/query.go](#)
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- [x/circuit/types/tx.pb.go](#)
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- [x/consensus/keeper/keeper.go](#)
- [x/consensus/keeper/keeper_test.go](#)
- [x/consensus/module.go](#)
- [x/consensus/testutil/expected_keepers_mock.go](#)
- [x/consensus/testutil/helpers.go](#)
- [x/consensus/types/codec.go](#)
- [x/consensus/types/keys.go](#)
- [x/consensus/types/msgs.go](#)
- [x/consensus/types/query.pb.go](#)
- [x/consensus/types/query.pb.gw.go](#)

- [x/consensus/types/tx.pb.go](#)
- [x/crisis/abci.go](#)
- [x/crisis/autocli.go](#)
- [x/crisis/exported/exported.go](#)
- [x/crisis/keeper/genesis.go](#)
- [x/crisis/keeper/genesis_test.go](#)
- [x/crisis/keeper/keeper.go](#)
- [x/crisis/keeper/keeper_test.go](#)
- [x/crisis/keeper/migrator.go](#)
- [x/crisis/keeper/msg_server.go](#)
- [x/crisis/keeper/msg_server_test.go](#)
- [x/crisis/migrations/v2/migrate.go](#)
- [x/crisis/migrations/v2/migrate_test.go](#)
- [x/crisis/module.go](#)
- [x/crisis/testutil/expected_keepers_mock.go](#)
- [x/crisis/types/codec.go](#)
- [x/crisis/types/errors.go](#)
- [x/crisis/types/events.go](#)
- [x/crisis/types/expected_keepers.go](#)
- [x/crisis/types/genesis.go](#)
- [x/crisis/types/genesis.pb.go](#)
- [x/crisis/types/keys.go](#)
- [x/crisis/types/legacy_params.go](#)
- [x/crisis/types/messages.go](#)
- [x/crisis/types/route.go](#)
- [x/crisis/types/tx.pb.go](#)
- [x/distribution/abci.go](#)
- [x/distribution/autocli.go](#)
- [x/distribution/client/cli/tx.go](#)
- [x/distribution/client/cli/tx_test.go](#)
- [x/distribution/client/cli/util_test.go](#)
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- [x/distribution/client/common/common_test.go](#)
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- [x/nft/testutil/expected_keepers_mock.go](#)
- [x/nft/tx.pb.go](#)
- [x/params/autocli.go](#)
- [x/params/client/cli/tx.go](#)
- [x/params/client/cli/tx_test.go](#)
- [x/params/client/proposal_handler.go](#)
- [x/params/client/utls/utls.go](#)
- [x/params/client/utls/utls_test.go](#)
- [x/params/doc.go](#)
- [x/params/keeper/common_test.go](#)
- [x/params/keeper/grpc_query.go](#)
- [x/params/keeper/grpc_query_test.go](#)
- [x/params/keeper/keeper.go](#)
- [x/params/keeper/keeper_test.go](#)
- [x/params/module.go](#)
- [x/params/proposal_handler.go](#)
- [x/params/proposal_handler_test.go](#)
- [x/params/simulation/operations.go](#)
- [x/params/simulation/operations_test.go](#)
- [x/params/simulation/proposals.go](#)
- [x/params/simulation/proposals_test.go](#)
- [x/params/testutil/staking_keeper_mock.go](#)
- [x/params/types/common_test.go](#)
- [x/params/types/consensus_params_legacy.go](#)
- [x/params/types/deref_test.go](#)
- [x/params/types/doc.go](#)
- [x/params/types/keys.go](#)
- [x/params/types/paramset.go](#)
- [x/params/types/proposal/codec.go](#)
- [x/params/types/proposal/errors.go](#)
- [x/params/types/proposal/keys.go](#)
- [x/params/types/proposal/params.pb.go](#)
- [x/params/types/proposal/proposal.go](#)
- [x/params/types/proposal/proposal_test.go](#)
- [x/params/types/proposal/query.pb.go](#)
- [x/params/types/proposal/query.pb.gw.go](#)
- [x/params/types/querier.go](#)
- [x/params/types/subspace.go](#)

- [x/params/types/subspace_test.go](#)
- [x/params/types/table.go](#)
- [x/params/types/table_test.go](#)
- [x/simulation/client/cli/flags.go](#)
- [x/simulation/doc.go](#)
- [x/simulation/event_stats.go](#)
- [x/simulation/expected_keepers.go](#)
- [x/simulation/log.go](#)
- [x/simulation/mock_cometbft.go](#)
- [x/simulation/operation.go](#)
- [x/simulation/params.go](#)
- [x/simulation/params_test.go](#)
- [x/simulation/simulate.go](#)
- [x/simulation/transition_matrix.go](#)
- [x/simulation/util.go](#)
- [x/slashing/abci.go](#)
- [x/slashing/abci_test.go](#)
- [x/slashing/app_test.go](#)
- [x/slashing/autocli.go](#)
- [x/slashing/exported/exported.go](#)
- [x/slashing/keeper/genesis.go](#)
- [x/slashing/keeper/genesis_test.go](#)
- [x/slashing/keeper/grpc_query.go](#)
- [x/slashing/keeper/grpc_query_test.go](#)
- [x/slashing/keeper/hooks.go](#)
- [x/slashing/keeper/hooks_test.go](#)
- [x/slashing/keeper/infractions.go](#)
- [x/slashing/keeper/keeper.go](#)
- [x/slashing/keeper/keeper_test.go](#)
- [x/slashing/keeper/migrations.go](#)
- [x/slashing/keeper/msg_server.go](#)
- [x/slashing/keeper/msg_server_test.go](#)
- [x/slashing/keeper/params.go](#)
- [x/slashing/keeper/signing_info.go](#)
- [x/slashing/keeper/signing_info_test.go](#)
- [x/slashing/keeper/slash_redelegation_test.go](#)
- [x/slashing/keeper/unjail.go](#)
- [x/slashing/migrations/v1/types.go](#)
- [x/slashing/migrations/v2/keys.go](#)
- [x/slashing/migrations/v2/store.go](#)
- [x/slashing/migrations/v2/store_test.go](#)
- [x/slashing/migrations/v3/migrate.go](#)
- [x/slashing/migrations/v3/migrator_test.go](#)
- [x/slashing/migrations/v4/keys.go](#)

- [x/slashing/migrations/v4/migrate.go](#)
- [x/slashing/migrations/v4/migrate_test.go](#)
- [x/slashing/module.go](#)
- [x/slashing/simulation/decoder.go](#)
- [x/slashing/simulation/decoder_test.go](#)
- [x/slashing/simulation/genesis.go](#)
- [x/slashing/simulation/genesis_test.go](#)
- [x/slashing/simulation/operations.go](#)
- [x/slashing/simulation/operations_test.go](#)
- [x/slashing/simulation/proposals.go](#)
- [x/slashing/simulation/proposals_test.go](#)
- [x/slashing/testutil/app_config.go](#)
- [x/slashing/testutil/expected_keepers_mock.go](#)
- [x/slashing/testutil/params.go](#)
- [x/slashing/testutil/val_init_helpers.go](#)
- [x/slashing/types/codec.go](#)
- [x/slashing/types/errors.go](#)
- [x/slashing/types/events.go](#)
- [x/slashing/types/expected_keepers.go](#)
- [x/slashing/types/genesis.go](#)
- [x/slashing/types/genesis.pb.go](#)
- [x/slashing/types/keys.go](#)
- [x/slashing/types/msg.go](#)
- [x/slashing/types/msg_test.go](#)
- [x/slashing/types/params.go](#)
- [x/slashing/types/params_legacy.go](#)
- [x/slashing/types/querier.go](#)
- [x/slashing/types/query.pb.go](#)
- [x/slashing/types/query.pb.gw.go](#)
- [x/slashing/types/signing_info.go](#)
- [x/slashing/types/slashing.pb.go](#)
- [x/slashing/types/tx.pb.go](#)
- [x/staking/app_test.go](#)
- [x/staking/autocli.go](#)
- [x/staking/bench_test.go](#)
- [x/staking/client/cli/flags.go](#)
- [x/staking/client/cli/tx.go](#)
- [x/staking/client/cli/tx_test.go](#)
- [x/staking/client/cli/utils.go](#)
- [x/staking/exported/exported.go](#)
- [x/staking/genesis.go](#)
- [x/staking/genesis_test.go](#)
- [x/staking/keeper/abci.go](#)
- [x/staking/keeper/alias_functions.go](#)

- [x/staking/keeper/delegation.go](#)
- [x/staking/keeper/delegation_test.go](#)
- [x/staking/keeper/genesis.go](#)
- [x/staking/keeper/grpc_query.go](#)
- [x/staking/keeper/grpc_query_test.go](#)
- [x/staking/keeper/historical_info.go](#)
- [x/staking/keeper/historical_info_test.go](#)
- [x/staking/keeper/invariants.go](#)
- [x/staking/keeper/keeper.go](#)
- [x/staking/keeper/keeper_test.go](#)
- [x/staking/keeper/migrations.go](#)
- [x/staking/keeper/msg_server.go](#)
- [x/staking/keeper/msg_server_test.go](#)
- [x/staking/keeper/messages.go](#)
- [x/staking/keeper/params.go](#)
- [x/staking/keeper/period_delegation.go](#)
- [x/staking/keeper/pool.go](#)
- [x/staking/keeper/power_reduction.go](#)
- [x/staking/keeper/power_reduction_test.go](#)
- [x/staking/keeper/query_utils.go](#)
- [x/staking/keeper/slash.go](#)
- [x/staking/keeper/slash_test.go](#)
- [x/staking/keeper/test_common.go](#)
- [x/staking/keeper/unbonding.go](#)
- [x/staking/keeper/unbonding_test.go](#)
- [x/staking/keeper/validator.go](#)
- [x/staking/keeper/validator_test.go](#)
- [x/staking/keeper/val_state_change.go](#)
- [x/staking/migrations/v1/types.go](#)
- [x/staking/migrations/v2/keys.go](#)
- [x/staking/migrations/v2/store.go](#)
- [x/staking/migrations/v2/store_test.go](#)
- [x/staking/migrations/v3/json.go](#)
- [x/staking/migrations/v3/json_test.go](#)
- [x/staking/migrations/v3/keys.go](#)
- [x/staking/migrations/v3/store.go](#)
- [x/staking/migrations/v3/store_test.go](#)
- [x/staking/migrations/v4/keys.go](#)
- [x/staking/migrations/v4/migrations_test.go](#)
- [x/staking/migrations/v4/store.go](#)
- [x/staking/migrations/v5/keys.go](#)
- [x/staking/migrations/v5/migrations_test.go](#)
- [x/staking/migrations/v5/store.go](#)
- [x/staking/module.go](#)

- [x/staking/module_test.go](#)
- [x/staking/simulation/common_test.go](#)
- [x/staking/simulation/decoder.go](#)
- [x/staking/simulation/decoder_test.go](#)
- [x/staking/simulation/genesis.go](#)
- [x/staking/simulation/genesis_test.go](#)
- [x/staking/simulation/operations.go](#)
- [x/staking/simulation/operations_test.go](#)
- [x/staking/simulation/proposals.go](#)
- [x/staking/simulation/proposals_test.go](#)
- [x/staking/testutil/app_config.go](#)
- [x/staking/testutil/cmt.go](#)
- [x/staking/testutil/expected_keepers_mock.go](#)
- [x/staking/testutil/helpers.go](#)
- [x/staking/testutil/validator.go](#)
- [x/staking/types/authz.go](#)
- [x/staking/types/authz.pb.go](#)
- [x/staking/types/authz_test.go](#)
- [x/staking/types/codec.go](#)
- [x/staking/types/commission.go](#)
- [x/staking/types/commission_test.go](#)
- [x/staking/types/data_test.go](#)
- [x/staking/types/delegation.go](#)
- [x/staking/types/delegation_test.go](#)
- [x/staking/types/errors.go](#)
- [x/staking/types/events.go](#)
- [x/staking/types/expected_keepers.go](#)
- [x/staking/types/exported.go](#)
- [x/staking/types/genesis.go](#)
- [x/staking/types/genesis.pb.go](#)
- [x/staking/types/historical_info.go](#)
- [x/staking/types/historical_info_test.go](#)
- [x/staking/types/hooks.go](#)
- [x/staking/types/keys.go](#)
- [x/staking/types/keys_test.go](#)
- [x/staking/types/msg.go](#)
- [x/staking/types/msg_test.go](#)
- [x/staking/types/params.go](#)
- [x/staking/types/params_legacy.go](#)
- [x/staking/types/params_test.go](#)
- [x/staking/types/period_delegation.go](#)
- [x/staking/types/pool.go](#)
- [x/staking/types/querier.go](#)
- [x/staking/types/query.pb.go](#)

- [x/staking/types/query.pb.gw.go](#)
- [x/staking/types/staking.pb.go](#)
- [x/staking/types/tx.pb.go](#)
- [x/staking/types/validator.go](#)
- [x/staking/types/validator_test.go](#)
- [x/tx/decode/adr027.go](#)
- [x/tx/decode/decode.go](#)
- [x/tx/decode/decode_test.go](#)
- [x/tx/decode/errors.go](#)
- [x/tx/decode/unknown.go](#)
- [x/tx/decode/unknown_test.go](#)
- [x/tx/internal/testpb/1.pulsar.go](#)
- [x/tx/internal/testpb/signers.pulsar.go](#)
- [x/tx/internal/testpb/unknownproto.pulsar.go](#)
- [x/tx/signing/aminojson/aminojson.go](#)
- [x/tx/signing/aminojson/aminojson_test.go](#)
- [x/tx/signing/aminojson/any.go](#)
- [x/tx/signing/aminojson/bench_test.go](#)
- [x/tx/signing/aminojson/encoder.go](#)
- [x/tx/signing/aminojson/internal/aminojsonpb/aminojson.pulsar.go](#)
- [x/tx/signing/aminojson/internal/testpb/test.pulsar.go](#)
- [x/tx/signing/aminojson/json_marshal.go](#)
- [x/tx/signing/aminojson/json_marshal_test.go](#)
- [x/tx/signing/aminojson/options.go](#)
- [x/tx/signing/aminojson/time.go](#)
- [x/tx/signing/context.go](#)
- [x/tx/signing/context_test.go](#)
- [x/tx/signing/directaux/direct_aux.go](#)
- [x/tx/signing/directaux/direct_aux_test.go](#)
- [x/tx/signing/direct/direct.go](#)
- [x/tx/signing/direct/direct_test.go](#)
- [x/tx/signing/handler_map.go](#)
- [x/tx/signing/handler_map_test.go](#)
- [x/tx/signing/signer_data.go](#)
- [x/tx/signing/sign_mode_handler.go](#)
- [x/tx/signing/std/handler_map.go](#)
- [x/tx/signing/testutil/util.go](#)
- [x/tx/signing/textual/any.go](#)
- [x/tx/signing/textual/any_test.go](#)
- [x/tx/signing/textual/bench_test.go](#)
- [x/tx/signing/textual/bool.go](#)
- [x/tx/signing/textual/bool_test.go](#)
- [x/tx/signing/textual/bytes.go](#)
- [x/tx/signing/textual/bytes_test.go](#)

- [x/tx/signing/textual/coins.go](#)
- [x/tx/signing/textual/coins_test.go](#)
- [x/tx/signing/textual/coin_test.go](#)
- [x/tx/signing/textual/dec.go](#)
- [x/tx/signing/textual/dec_test.go](#)
- [x/tx/signing/textual/duration.go](#)
- [x/tx/signing/textual/duration_test.go](#)
- [x/tx/signing/textual/e2e_test.go](#)
- [x/tx/signing/textual/encode.go](#)
- [x/tx/signing/textual/encode_test.go](#)
- [x/tx/signing/textual/enum.go](#)
- [x/tx/signing/textual/enum_test.go](#)
- [x/tx/signing/textual/fuzz_test.go](#)
- [x/tx/signing/textual/handler.go](#)
- [x/tx/signing/textual/handler_test.go](#)
- [x/tx/signing/textual/internal/cbor/cbor.go](#)
- [x/tx/signing/textual/internal/cbor/cbor_test.go](#)
- [x/tx/signing/textual/internal/textualpb/doc.go](#)
- [x/tx/signing/textual/internal/textualpb/textual.pulsar.go](#)
- [x/tx/signing/textual/int.go](#)
- [x/tx/signing/textual/int_test.go](#)
- [x/tx/signing/textual/message.go](#)
- [x/tx/signing/textual/message_test.go](#)
- [x/tx/signing/textual/protorelect_list_test.go](#)
- [x/tx/signing/textual/repeated_test.go](#)
- [x/tx/signing/textual/string.go](#)
- [x/tx/signing/textual/string_test.go](#)
- [x/tx/signing/textual/timestamp.go](#)
- [x/tx/signing/textual/timestamp_test.go](#)
- [x/tx/signing/textual/tx.go](#)
- [x/tx/signing/textual/tx_test.go](#)
- [x/tx/signing/textual/types.go](#)
- [x/tx/signing/tx_data.go](#)
- [x/upgrade/abci.go](#)
- [x/upgrade/abci_test.go](#)
- [x/upgrade/autocli.go](#)
- [x/upgrade/client/cli/parse.go](#)
- [x/upgrade/client/cli/parse_test.go](#)
- [x/upgrade/client/cli/tx.go](#)
- [x/upgrade/doc.go](#)
- [x/upgrade/exported/exported.go](#)
- [x/upgrade/internal/conv/doc.go](#)
- [x/upgrade/internal/conv/string.go](#)
- [x/upgrade/internal/conv/string_test.go](#)

- [x/upgrade/keeper/grpc_query.go](#)
- [x/upgrade/keeper/grpc_query_test.go](#)
- [x/upgrade/keeper/keeper.go](#)
- [x/upgrade/keeper/keeper_test.go](#)
- [x/upgrade/keeper/migrations.go](#)
- [x/upgrade/keeper/migrations_test.go](#)
- [x/upgrade/keeper/msg_server.go](#)
- [x/upgrade/keeper/msg_server_test.go](#)
- [x/upgrade/module.go](#)
- [x/upgrade/plan/downloader.go](#)
- [x/upgrade/plan/downloader_test.go](#)
- [x/upgrade/plan/info.go](#)
- [x/upgrade/plan/info_test.go](#)
- [x/upgrade/types/codec.go](#)
- [x/upgrade/types/errors.go](#)
- [x/upgrade/types/handler.go](#)
- [x/upgrade/types/keys.go](#)
- [x/upgrade/types/plan.go](#)
- [x/upgrade/types/plan_test.go](#)
- [x/upgrade/types/proposal.go](#)
- [x/upgrade/types/proposal_test.go](#)
- [x/upgrade/types/querier.go](#)
- [x/upgrade/types/query.pb.go](#)
- [x/upgrade/types/query.pb.gw.go](#)
- [x/upgrade/types/storeloader.go](#)
- [x/upgrade/types/storeloader_test.go](#)
- [x/upgrade/types/tx.pb.go](#)
- [x/upgrade/types/upgrade.pb.go](#)

Out-of-Scope: Third party dependencies and economic attacks.

REMEDATION COMMIT ID:



- <https://github.com/piplabs/cosmos-sdk/pull/28/commits/123577ba125bbd0099e18c53bee4645f8cdd254b>
- <https://github.com/piplabs/cosmos-sdk/pull/31/commits/f8dcf3924e9ca5ceac8c72debc1bf606b20da697>

Out-of-Scope: New features/implementations after the remediation commit IDs.

6. ASSESSMENT SUMMARY & FINDINGS OVERVIEW

CRITICAL

0

HIGH

1

MEDIUM

0

LOW

1

INFORMATIONAL

0

SECURITY ANALYSIS	RISK LEVEL	REMEDIATION DATE
KEY TYPO MAY ALLOW STORE CORRUPTION	HIGH	SOLVED - 11/27/2024
INCORRECT DEFAULT PERIOD DURATION	LOW	SOLVED - 12/09/2024

7. FINDINGS & TECH DETAILS

7.1 KEY TYPO MAY ALLOW STORE CORRUPTION

// HIGH

Description

There are two keys with the same value. As they are used as the prefix to retrieve the values under the Cosmos store, then it is possible to corrupt each other by being used on completely different parts of the codebase, as they would be querying the same key underneath.

The keys for the store prefixes are defined as follows:

<https://github.com/piplabs/cosmos-sdk/blob/c553dc6734da1d49fcde55efd825d84998051dd7/x/staking/types/keys.go#L28C1-L60C2>

```
var (
    // Keys for store prefixes
    // Last* values are constant during a block.
    LastValidatorPowerKey = []byte{0x11} // prefix for each key to a validator
    LastTotalPowerKey    = []byte{0x12} // prefix for the total power

    ValidatorsKey          = []byte{0x21} // prefix for each key to a validator
    ValidatorsByConsAddrKey = []byte{0x22} // prefix for each key to a validator
    ValidatorsByPowerIndexKey = []byte{0x23} // prefix for each key to a validator

    DelegationKey          = []byte{0x31} // key for a delegation
    UnbondingDelegationKey = []byte{0x32} // key for an unbonding-delegation
    UnbondingDelegationByValIndexKey = []byte{0x33} // prefix for each key for an unbonding-delegation
    RedelegationKey        = []byte{0x34} // key for a redelegation
    RedelegationByValSrcIndexKey = []byte{0x35} // prefix for each key for a redelegation
    RedelegationByValDstIndexKey = []byte{0x36} // prefix for each key for a redelegation
    PeriodDelegationKey     = []byte{0x37} // key for a period delegation

    UnbondingIDKey    = []byte{0x37} // key for the counter for the incremental unbonding-delegation
    UnbondingIndexKey = []byte{0x38} // prefix for an index for looking up unbonding-delegations
    UnbondingTypeKey  = []byte{0x39} // prefix for an index containing the type of unbonding-delegation

    UnbondingQueueKey    = []byte{0x41} // prefix for the timestamps in unbonding-delegations
    RedelegationQueueKey = []byte{0x42} // prefix for the timestamps in redelegations
    ValidatorQueueKey    = []byte{0x43} // prefix for the timestamps in validators
```

```

HistoricalInfoKey    = []byte{0x50} // prefix for the historical info
ValidatorUpdatesKey = []byte{0x61} // prefix for the end block validator u

ParamsKey = []byte{0x51} // prefix for parameters for module x/staking

DelegationByValIndexKey = []byte{0x71} // key for delegations by a validator
)

```

However, in the case of `PeriodDelegationKey` and `UnbondingIDKey` keys:

<https://github.com/piplabs/cosmos-sdk/blob/c553dc6734da1d49fcde55efd825d84998051dd7/x/staking/types/keys.go#L44C1-L46C105>

```

PeriodDelegationKey      = []byte{0x37} // key for a period delegation
UnbondingIDKey           = []byte{0x37} // key for the counter for the incremental

```

They share the same value, which allows for corruption of each other values from completely different parts of the codebase.

BVSS

AO:A/AC:L/AX:L/C:N/I:M/A:L/D:N/Y:N/R:N/S:C (7.6)

Recommendation

Change one of them to another value not being used by other keys.

Remediation

SOLVED: Changed the key `UnbondingIDKey` to `0x40`.

Remediation Hash

<https://github.com/piplabs/cosmos-sdk/pull/28/commits/123577ba125bbd0099e18c53bee4645f8cdd254b>

7.2 INCORRECT DEFAULT PERIOD DURATION

// LOW

Description

The default periods have a flaw where the period of type **1** expects **Duration** of **3 months**, but instead, it receives a fixed duration of **30 days**.

The definition is wrong:

<https://github.com/piplabs/cosmos-sdk/blob/c553dc6734da1d49fcde55efd825d84998051dd7/x/staking/types/params.go#L44C1-L65C2>

```
var DefaultPeriods = []Period{
{
    PeriodType:      0,
    Duration:         time.Duration(0),
    RewardsMultiplier: math.LegacyOneDec(), // 1
},
{
    PeriodType:      1,
    Duration:         time.Hour * 24 * 30,           // 3 months
    RewardsMultiplier: math.LegacyNewDecWithPrec(1051, 3), // 1.051
},
{
    PeriodType:      2,
    Duration:         time.Hour * 24 * 365,          // 1 year
    RewardsMultiplier: math.LegacyNewDecWithPrec(116, 2), // 1.16
},
{
    PeriodType:      3,
    Duration:         time.Hour * 24 * 30 * 18,       // 18 months
    RewardsMultiplier: math.LegacyNewDecWithPrec(134, 2), // 1.34
},
}
```

The second one is wrong, as it receives the wrong duration:

<https://github.com/piplabs/cosmos-sdk/blob/c553dc6734da1d49fcde55efd825d84998051dd7/x/staking/types/params.go#L50C1-L54C4>


```
{
  PeriodType:      1,
  Duration:        time.Hour * 24 * 30,           // 3 months
  RewardsMultiplier: math.LegacyNewDecWithPrec(1051, 3), // 1.051
},
```

BVSS

AO:A/AC:L/AX:L/C:N/I:N/A:L/D:N/Y:N/R:N/S:C (3.1)

Recommendation

Change the code to:

```
{
  PeriodType:      1,
  Duration:        time.Hour * 24 * 30 * 3,       // 3 months
  RewardsMultiplier: math.LegacyNewDecWithPrec(1051, 3), // 1.051
},
```

Remediation

SOLVED: Fixed by updating the given durations to its correct values.

Remediation Hash

<https://github.com/piplabs/cosmos-sdk/pull/31/commits/f8dcf3924e9ca5ceac8c72debc1bf606b20da697>

Halborn strongly recommends conducting a follow-up assessment of the project either within six months or immediately following any material changes to the codebase, whichever comes first. This approach is crucial for maintaining the project's integrity and addressing potential vulnerabilities introduced by code modifications.